

*Teaching Notes 4A**

Discount Rates

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Overview

- Earlier, we focused on the numerator of the NPV formula: Cash flows
 - Now, we focus on the denominator: Discount rate
- Two determinants of the discount rate for a cash flow at a given time horizon:
 1. Time value of money
 2. Risk

Objectives

- Understand the discount rate that captures the “business risk”
- Understand the relationship between the discount rate that captures business risk and the discount rate promised to debt and equity holders
 - The discount rate that captures returns to equity holders captures both business risk and financial risk
- Learn how to estimate the discount rate that captures business risk

Assumptions

- Start by calculating the discount rate for a project
 - r_A : captures “**business risk**” of the project
Called the discount rate of a “**100% equity financed**” firm with the same business risk
- For now, assume r_A doesn’t depend on how you finance that project
- Discount rate for the project is the market return of an “equally risky” security
 - Opportunity cost of capital: Imagine investing in an equally risky security. Then returns on the project and the security should be the same.

Risk, Asset Pricing, and Discount Rates

- Discount rate = r_f + adjustment for risk
- For what types of risk do investors require higher expected returns?
 - Investors care about risk that is correlated with the market, i.e. systematic risk
 - Don't care about idiosyncratic (diversifiable) risk
- Just as you ask this question when buying and selling financial securities, you must also do so when investing in projects

Flashback to Earlier “Stuff”

- The CAPM states

$$r_i = r_f + \beta_i \times (r_m - r_f)$$

- r_i = required asset return
- r_f = today's risk-free rate with the same duration as the asset
- r_m = required return on the market
- $(r_m - r_f)$ = Market risk premium
- β_i = sensitivity of the asset return to market movements

$$\beta_i = \frac{Cov(r_i, r_m)}{Var(r_m)}$$

Equity Betas and Asset Betas

- The context for the CAPM that we discussed earlier was generally equity investing
 - Beta was an **equity beta (β_E)**, the sensitivity of a stock's return to the return on the market
- The context now is investing in a machine, plant, technology, or firm that produces cash flows
 - Need a different kind of beta, an **asset beta (β_A)**

Finding the Asset Beta

- To apply the formula, we must use an asset beta that is appropriate to the specific cash flows under consideration
- One way is to find a traded firm with the same risk as our project and use its beta (the “Twin Method”)
- But for a traded firm, we only see its equity beta
 - Can calculate the asset beta from the equity beta
 - Will also need to think about leverage

The “Twin Method”

1. Identify a comparable twin firm
 - A firm with activities that are equally risky as our project
2. Look up (or calculate) its equity beta
3. Unlever the equity beta to get the asset beta
 - Go from equity beta of “twin” firm to asset beta of “twin firm”
4. Asset beta of twin firm = Asset beta of project
5. The asset beta goes into the CAPM formula to determine the discount rate

Unlevering Equity Beta of “Twin Firm”

- To calculate the asset beta from the equity beta, first recall the equality

$$A = E + D$$

- We can consider the assets of a firm as a portfolio of its liabilities: equity and debt. Therefore the company's total asset return must be

$$r_A = r_E \times \left(\frac{E}{E + D} \right) + r_D \times \left(\frac{D}{E + D} \right)$$

- That is, the total return (r_A) is divided between equity (E) and debt holders (D):
 - Equity holders receive r_E
 - Debt holders receive r_D
 - E and D are the market values of equity and debt

Unlevering Equity Beta of “Twin Firm”

- Covariance is linear

$$Cov(r_M, r_A) = Cov(r_M, r_E) \times \left(\frac{E}{E + D} \right) + Cov(r_M, r_D) \times \left(\frac{D}{E + D} \right)$$

- Dividing by $var(r_M)$ gives

$$\frac{Cov(r_M, r_A)}{Var(r_M)} = \frac{Cov(r_M, r_E)}{Var(r_M)} \times \left(\frac{E}{E + D} \right) + \frac{Cov(r_M, r_D)}{Var(r_M)} \times \left(\frac{D}{E + D} \right)$$

Unlevering Formulas

- In other words,

$$\beta_A = \beta_E \times \left(\frac{E}{E + D} \right) + \beta_D \times \left(\frac{D}{E + D} \right)$$

- Or alternatively,

$$\beta_E = \beta_A + \left(\frac{D}{E} \right) \times (\beta_A - \beta_D)$$

- Note: learn these formulas. We will keep coming back to them

Again, Why Not Use Twin's β_E ?

- Note the problem:

$$\beta_E = \underbrace{\beta_A}_{\text{Business Risk}} + \left(\frac{D}{E}\right) \times \underbrace{(\beta_A - \beta_D)}_{\text{Financial Risk}}$$

- **Financial risk** is the risk that equity holders must bear in the presence of higher-priority debt (not bankruptcy risk)

What About Debt Beta?

- If there is little debt in the firm (D/E low) and debt is “safe”, then assume $\beta_D = 0$.
- If there is more debt, then use the average beta of debt for its particular rating class.
 - Fama and French (1993) estimates are provided on the next page and serve as typical examples.

Fama-French (1993) Debt Betas

Type of Bond	Estimated β
Government maturity 1- 5 years	0.08
Government maturity 6- 10 years	0.13
Corporate Aaa	0.20
Corporate Aa	0.20
Corporate A	0.21
Corporate Baa	0.23
Corporate lower grade	0.31

Can We Use Firm's Own Asset Beta?

- Are there any circumstances under which the implementing firm's asset beta will be the correct asset beta for the new project?
- Only when the project's characteristics are identical to the rest of the firm
 - I.e. an expansion of operations
- But, in some cases, the best we can do is to make this assumption

Note: Beta For a Portfolio

- We used the fact that beta for a firm's assets is a weighted average of the debt and equity betas
- The same thing holds on the asset side. If you have two assets (A_1 and A_2) so that the total is $A = A_1 + A_2$, then

$$\beta_A = \beta_1 \times \left(\frac{A_1}{A} \right) + \beta_2 \times \left(\frac{A_2}{A} \right)$$

- This can be useful, i.e., if you want to unlever a company with distinct operations or cash
 - Also, this formula can be easily generalized for a case with more than two assets.

Example 1: Risk-Free Rate

Trump Calls for a Big Fed Rate Cut, Again Criticizes Central Bank Chairman

President says Jerome Powell has 'horrendous lack of vision,' says big reduction would improve economy

By *Rebecca Ballhaus*, *Andrew Restuccia* and *Paul Kiernan*

Updated Aug. 19, 2019 7:49 pm ET

WASHINGTON—President Trump on Monday called for the Federal Reserve to sharply cut interest rates and again criticized the central bank's chairman for a "horrendous lack of vision," while reiterating his belief that the U.S. economy is strong.

- Why does Trump want the Fed to cut interest rates?
 - Because stock prices would rise!

Example 2: Mobile Phones

- Telecommunications conglomerate AMCI wants to produce cell phones
 - It will use 100% equity financing
- Let's say another firm, Nokorola, only produces mobile phones
 - The market value of Nokorola's equity is $E = \$1.2\text{M}$, with a beta of 1.2
 - The book value of Nokorola's debt is $D = \$0.4\text{M}$, and it has a beta of 0.4
- Suppose the risk-free rate is 10% and the average return on the market is 15%
- What discount rate is appropriate for AMCI's mobile phone project?

Example 3: Leveraged Recap

- More levering and unlevering arithmetic
- Suppose a firm issues debt to repurchase its own equity
- Assume this does not change the performance of the firm's assets
- Then we can infer the debt beta by looking at how the equity changes

Leveraged Recap Example (cont.)

- The firm's debt goes from 20% to 50% of total assets
- Say the debt beta is zero initially
- The equity beta is available from market data. It is 0.75 before the recap, and 0.90 after
- Let's find the debt beta after the recap

Inferring the New β_D

- The starting point is to find the β_A

$$\beta_A = \frac{D_{pre}}{D_{pre} + E_{pre}} \times 0 + \frac{E_{pre}}{D_{pre} + E_{pre}} \times 0.75 = 0.60$$

- Then assume $\beta_{A,pre} = \beta_{A,post} = \beta_A$

$$\begin{aligned}\beta_{E,post} &= \beta_A + \frac{D_{post}}{E_{post}} (\beta_A - \beta_{D,post}) \\ \Rightarrow \beta_{D,post} &= 0.30\end{aligned}$$

Takeaways

- Two ingredients to getting NPV right
 - Cash Flows
 - Discount Rates
- The right discount rate should be PROJECT specific and depends on project's ASSET beta
- Find an asset beta using twins and unlever their equity betas. Once we have the asset beta:
 - Can plug into the CAPM to find $r_A = r_f + \beta_A^*(r_m - r_f)$
 - r_A = the right discount rate to use if the project is 100% equity financed